Md Aminul Islam

Electronic Information Engineering | IoT & Embedded Systems | Information science

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RESEARCH STATEMENT & INTERESTS

I am a motivated engineer with a strong foundation in electronic information engineering and hands-on expertise in Internet of Things (IoT) systems, embedded computing, machine learning integration, and cloud computing. My research interests centre on intelligent IoT architectures, smart energy systems, edge-AI applications, and cloud-based distributed systems. I am seeking a PhD position in England where I can contribute to and advance research at the intersection of IoT, embedded systems, applied machine learning, and cloud computing infrastructure.

EDUCATION

Master of Science — Information Science
2024 – Present

Trine University, Michigan, USA

- GPA: 3.66 / 4.0 (in progress)
- Research focus: Cloud computing architectures, distributed systems, and cloud-IoT integration.
- Core modules: Cloud infrastructure, data management, information systems, and distributed computing.

Bachelor of Engineering — Electronic Information Engineering
Sept 2018 – Jun 2022

Yunnan University, Kunming, China

- GPA: 3.27 / 4.0
- Thesis: Smart Charging Station — designed and implemented an IoT-enabled EV charging management system incorporating real-time monitoring, load balancing, and mobile user interface.

Diploma in Engineering — Electrical Engineering
2014– 2018

Rajshahi Polytechnic Institute, Rajshahi, Bangladesh

- GPA: 3.31 / 4.0 (Distinction)
- Core modules: Power systems, PLC programming, circuit board design, and industrial automation.

RESEARCH & TECHNICAL PROJECTS

Smart Charging Station (Bachelor Thesis)
2021– 2022

- Developed a full-stack IoT solution for electric vehicle charging infrastructure, integrating ESP32 microcontrollers with cloud-based data logging and a mobile application, supporting up to 10 simultaneous charging sessions.
- Implemented real-time energy monitoring and user authentication across 5+ sensor nodes, achieving under 500ms system response time and demonstrating end-to-end system design capability.

Embedded IoT Systems — Independent & Freelance Projects
2019 – Present

- Designed and deployed 10+ sensor-driven automation projects using Raspberry Pi, Arduino, and ESP32 platforms for clients across Bangladesh and China.
- Developed prototypes involving environmental monitoring, home automation, and low-power wireless communication (MQTT, Wi-Fi, Bluetooth), reducing manual monitoring effort by an estimated 60%.
- Applied TensorFlow.js for lightweight on-device inference in IoT contexts, achieving real-time classification on resource-constrained hardware.

PLC & Industrial Automation
2017 – 2018

- Completed projects involving Programmable Logic Controller (PLC) and Programmable Logic Device (PLD) programming for industrial process control.
- Gained practical exposure to circuit board design and prototyping.

PROFESSIONAL EXPERIENCE

Electronic Engineer *2019 – 2021*

Various engineering firms, Bangladesh & China

- Worked across two countries in technical and client-facing roles, developing expertise in engineering solutions delivery and cross-cultural communication.
- Managed technical sales cycles, communicated complex product specifications to clients, and supported post-sales technical implementation.

Chinese–English Technical Translator *2020 – 2021*

Xiaomi Mobile Company, Bangladesh

- Provided Chinese–English translation in professional and technical settings, facilitating communication between Bangladeshi and Chinese counterparts.

Teacher — English Language & Math *2019 – 2021*

Private Instruction, Bangladesh

- Delivered language and technical instruction to colleagues and students, supporting development of English language proficiency and core Math.

TECHNICAL SKILLS

Programming & Software

- Languages: Python, C++, JavaScript
- Machine Learning / AI: TensorFlow.js, applied ML model deployment on edge devices
- Version control: Git; Database: MySQL
- Networking & infrastructure: Cisco fundamentals

Hardware & Systems Engineering

- Microcontrollers & SBCs: Arduino, ESP32, Raspberry Pi
- Industrial systems: PLC, PLD programming; circuit board design and prototyping
- IoT protocols: MQTT, HTTP, Wi-Fi, Bluetooth

Design & Productivity Tools

- 3D modelling tools (mechanical and electronic enclosure design)
- Adobe Photoshop, Adobe Illustrator; Microsoft Office Suite
- Computer graphics and technical illustration

Publications

English (Professional working proficiency)

LANGUAGES

English (Professional working proficiency) • **Chinese (Mandarin)** (Professional working proficiency) • **Bengali** (Native) • **Hindi** (Professional working proficiency) • **Arabic** (Conversational)

COMMUNITY ENGAGEMENT & VOLUNTEERING

- Founder, community charity foundation — established and managed a local charitable organization in my hometown focused on social welfare and community development.
- Regular blood donor (twice annually), supporting national health initiatives.

REFERENCES

Available upon request. Academic and professional referees can be provided from Yunnan University (thesis supervisor) and former employers in Bangladesh and China.